PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Lower-Bound Guarantees for Resilient Power Transformer Thermal Dynamics Forecasting

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Abstract

Forecasting power transformer thermal dynamics is critical for predictive maintenance, dynamic rating, and asset protection. While Kolmogorov-Arnold Networks (KANs) offer expressive functional representation, dynamic B-spline grid adaptations cause severe training instability—the Grid Update Dilemma—while their long-horizon stability remains vulnerable. This paper introduces PSO-TCWavKAN, a unified architecture integrating continuous dyadic Morlet wavelet filter banks with multi-kernel trend-seasonal decomposition and an instance-adaptive neural routing gate. The routing gate provides asymptotic linear structural recovery (Theorem 1), mathematically proving that the hypothesis space topological closure contains canonical DLinear (K1=K2=25) as gating weights close, retaining linear fidelity during high-variance transients. A Comprehensive Learning Particle Swarm Optimizer (CLPSO) with Clerc constriction factor (chi = 0.7298) automates hyperparameter warm-start across a bounded 4-D parameter space in 25 evaluations, bypassing non-convex ripple traps prior to AdamW fine-tuning and Stochastic Weight Averaging (SWA). Evaluated across 16 canonical configurations on Electricity Transformer Temperature (ETT) telemetry, PSO-TCWavKAN achieves the lowest Mean Squared Error in 7 of 8 lookback-horizon configurations on volatile ETTh2, reducing MSE versus DLinear by up to 13.6% (and up to 8.5% for H <= 336), with nominal uncorrected improvement over NLinear ($p = 0.0386$; Holm-corrected $p_{Holm} = 0.116$). With 44.8 K parameters (3.8x reduction over PatchTST Lite, 169.6 K) and 3.4 ms inference latency, the model establishes a compact, highly reliable paradigm for edge SCADA telemetry monitoring.

Keywords: Kolmogorov-Arnold Networks; Wavelet Filter Bank; Particle Swarm Optimization; Transformer Temperature Forecasting; Time-Series Forecasting; Reversible Instance Normalization

1. Introduction

High-voltage electrical power transformers represent the critical capital assets of electrical energy transmission and distribution grids. In modern utility networks, continuous operational loading induces substantial Joule losses in winding conductors and core eddy-current dissipation, generating intense internal heat. The thermal state of a transformer is conventionally characterized by two primary physical indicators: the top-oil temperature (TOT or OT) and the internal winding hot-spot temperature (HST). Operating temperatures are directly coupled to the irreversible degradation of solid cellulose insulation paper and dielectric mineral oil via Arrhenius thermal aging kinetics. In accordance with IEEE Std C57.91 [21], IEC 60076-7 [22], and the newly established IEEE Std C57.165-2024 guide for transformer temperature measurements [47], every sustained 6 °C elevation of the winding hot-spot temperature above the 110 °C reference approximately doubles the insulation aging acceleration rate, markedly escalating the probability of catastrophic dielectric breakdown and grid-level outages [25, 26, 48]. Consequently, accurate multi-step-ahead thermal forecasting is paramount for condition-based maintenance, Dynamic Thermal Rating (DTR) of transmission corridors, and automatic supervisory activation of substation forced-cooling equipment [23, 49].

In field substation telemetry, such as the canonical Electricity Transformer Temperature (ETT) benchmark established by Zhou et al. [3], supervisory monitoring relies on Resistance Temperature Detectors (RTDs) positioned in top-oil thermowells, recording Top-Oil Temperature (OT). While winding hot-spot temperature (HST) dictates insulation loss-of-life, direct HST measurement requires down-hole optical fiber sensors that are seldom retrofitted to in-service transformer fleets. Instead, utilities monitor OT and use standardized differential thermal balance equations (IEEE C57.91 / IEC 60076-7) to compute HST dynamically from predicted OT, ambient temperature, and load currents. Accurate multi-step OT prediction therefore serves as the indispensable thermodynamic boundary condition for real-time asset protection and DTR calculations [11, 48].

Following the release of the canonical ETT benchmark, sequential thermal modeling transitioned through multiple paradigms, progressing from Recurrent Neural Networks to Transformer architectures equipped with sparse or linear self-attention mechanisms, including Informer [3], Autoformer [14], and FEDformer [15], as surveyed extensively by Tay et al. [38]. However, this trajectory was dramatically disrupted by Zeng et al. [4], who demonstrated that simple linear mapping layers with moving-average trend decomposition (DLinear, NLinear) frequently outperform complex Transformer

architectures across standard benchmarks—a phenomenon widely recognized as the linear model dilemma. While patched Transformers such as PatchTST [5] and inverted token architectures like iTransformer [16] achieved notable gains by enforcing channel-independence and cross-variate correlations, their multi-megabyte parameter scales and high computational latency impede deployment on resource-constrained substation Remote Terminal Units (RTUs) and edge SCADA gateways [45, 46].

Concurrently, the scientific machine learning community witnessed the emergence of Kolmogorov-Arnold Networks (KANs), introduced by Liu et al. [1]. Grounded in the Kolmogorov-Arnold representation theorem, KANs place learnable univariate B-spline functions on network edges rather than fixed non-linear activations at nodes, yielding superior functional expressiveness, enhanced parameter efficiency, and inherent interpretability [6, 10, 44]. To address the limitations of piecewise splines, recent studies have explored alternative orthogonal expansions, including wavelet activations (Wav-KAN [2]), Chebyshev polynomials (ChebyKAN [43]), and structural dropout regularizers (DropKAN [42]), with applications extending to engineering dynamics and hybrid forecasting (HAKformer [12]).

Nevertheless, the direct application of classical KAN architectures to complex industrial electrical telemetry exposes four severe failure modes: (1) The Grid Update Dilemma: Standard B-spline KANs rely on discrete knot re-meshing during training to cover shifting activation distributions. As proven theoretically and demonstrated in our diagnostic tracking, abrupt grid reallocation alters the underlying coordinate manifold, invalidating optimizer moment vectors (m_t, v_t) in AdamW and triggering severe validation loss spikes and training divergence [44]. (2) Lack of Multi-Scale Spectral Bandpass Localization: Uniform spline bases lack explicit multi-octave bandpass selectivity. Power transformer thermal sequences reflect the physical superposition of slow bulk oil thermal inertia (tau_oil approx 1.5 - 3.0 hours) and rapid winding transients (tau_wind approx 4 - 10 minutes) [21, 22]. Without multi-scale localization, unconstrained splines suffer from spectral dispersion and high-frequency noise sensitivity. (3) The Linear Model Performance Deficit: Across canonical horizons (96 to 720 steps), unconstrained non-linear deep models frequently underperform simple linear models on trend-dominated sequences through error accumulation and non-stationary overfitting [4]. (4) Non-Convex Wavelet Ripple Traps: Introducing oscillatory functional bases into network edges creates highly non-convex loss surfaces with dense local minima, causing standard gradient descent to stagnate in suboptimal basins [17, 18].

To systematically resolve these challenges without compromise, this paper introduces PSO-TCWavKAN, a swarm-optimized temporal convolutional wavelet Kolmogorov-Arnold network engineered for resilient power transformer thermal dynamics forecasting. The core scientific contributions of this investigation are fivefold:

1. Theoretical Resolution of the KAN Grid Update Dilemma: We replace discrete B-spline grids with a continuous, parameterized dyadic Morlet wavelet filter bank across K_w = 5 learnable octaves. Because Morlet wavelets are infinitely differentiable (C^infinity) over R, basis frequencies and scale parameters are optimized continuously via gradient descent, eliminating discrete knot reallocation entirely and guaranteeing stable optimizer moment progression by construction.

2. Multiscale Localized Time-Frequency Representation: Tailored to the physical dual-timescale thermodynamics of mineral-oil-immersed transformers, our continuous dyadic Morlet filter bank achieves the theoretical Heisenberg-Gabor lower bound (Delta t * Delta omega = 1/2). Lower octaves capture bulk oil thermal capacitance and diurnal load cycles, while higher octaves capture abrupt industrial load switching and transient thermal excursions, overcoming MLP spectral bias.

3. Empirical Superiority in High-Volatility Regimes: On the high-volatility ETTh2 benchmark, PSO-TCWavKAN achieves the lowest MSE in 7 out of 8 lookback-horizon configurations among all competing models, reducing MSE versus DLinear by up to 13.6% at H=720. Under Theorem 1 (Asymptotic Linear Structural Recovery), we prove that the topological closure of the model hypothesis space contains canonical DLinear (K1=K2=25), preventing linear model collapse while dynamically routing volatile non-linear residuals through the wavelet pathway.

4. Computational and Parameter Efficiency for Edge Grid Monitoring: With only 44,841 parameters for the base configuration (L=96, H=96), PSO-TCWavKAN achieves a 3.8x parameter reduction over PatchTST Lite (169.6 K) and trains 3.3x faster than standard B-spline KANs (1.24 s/epoch on an NVIDIA Tesla T4 GPU). Its ultra-low batch inference latency of 3.4 ms satisfies the real-time operational requirements of substation SCADA RTUs and IEEE C57.165-2024 monitoring units.

5. Automated Algorithmic Co-Design via Bounded CLPSO: We implement a constriction-factor Comprehensive Learning Particle Swarm Optimizer (chi = 0.7298, c1=c2=2.05) across a 4-D continuous parameter space (omega_0, q, log_alpha, log_10(eta)). With a strictly bounded budget of exactly 25 surrogate evaluations (5 particles x 5 generations), CLPSO reliably localizes the low-risk global basin prior to AdamW fine-tuning and Stochastic Weight Averaging (SWA), eliminating manual hyperparameter guesswork.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the PSO-TCWavKAN architecture and Theorem 1. Section 4 presents the experimental setup. Sections 5 through 8 present benchmark evaluations,

ablation studies, and computational efficiency profiling. Section 9 discusses engineering implications, and Section 10 concludes.

2. Related Work

2.1. Power Transformer Thermal Dynamics and Operational Standards

Thermal modeling of mineral-oil-immersed power transformers has historically relied on lumped-parameter differential equations standardized in IEEE Std C57.91 [21] and IEC 60076-7 [22]. Seminal physical models—such as the thermal-electrical analogy proposed by Susa et al. [23], the fundamental thermal approaches by Swift et al. [24], and dynamic hot-spot calculation methods by Radakovic and Feser [25]—estimate top-oil temperature (OT) and winding hot-spot temperature (HST) from ambient temperature, wind velocity, and load currents [26]. However, standard differential equations require precise hydrodynamic parameters (oil viscosity, winding resistances, cooling dissipation coefficients) that drift significantly over operational lifetimes [48]. The newly ratified IEEE Std C57.165-2024 [47] emphasizes the integration of continuous telemetry monitoring to support Dynamic Thermal Rating (DTR) and mitigate extreme thermal risks [49]. In a pioneering study published in Electrical Engineering, Bezerra et al. [11] demonstrated that combining second-generation wavelets with neural networks captures non-stationary oil temperature dynamics superior to classical thermal equations. Nevertheless, their approach relied on discrete lifting wavelets prone to boundary artifacts and standard MLPs lacking structural linear recovery guarantees or automated hyperparameter optimization.

2.2. Non-Stationary Time-Series Forecasting: Transformers to Linear Baselines

Following the introduction of Informer by Zhou et al. [3], Transformer variants integrated specialized temporal mechanisms, such as Auto-Correlation in Autoformer [14] and frequency-domain representations in FEDformer [15]. PatchTST [5] demonstrated that segmenting univariate series into subseries patches significantly enhances temporal locality while reducing self-attention complexity. Concurrently, Zeng et al. [4] revealed that simple single-layer linear models with moving-average trend decomposition (DLinear, NLinear) frequently outperform complex self-attention architectures across long-term forecasting benchmarks, exposing the vulnerability of unconstrained attention to temporal distribution shifts. Recent advances, including iTransformer [16] (variate-centric inverted tokens) and ModernTCN [45] (pure depthwise convolution), have established competitive baselines, while foundation models such as Time-LLM [46] demonstrated zero-shot transferability at the expense of billions of parameters. To address non-stationarity, Kim et al. [13] introduced Reversible Instance Normalization (RevIN) to remove mean and variance shifts, establishing an indispensable pre-processing foundation for sequential deep models.

2.3. Kolmogorov-Arnold Networks and Wavelet Orthogonal Bases

Grounded in the Kolmogorov-Arnold representation theorem, KANs [1] place learnable univariate functions on network edges rather than fixed nodal activations. The foundational formulation utilized B-splines; however, knot re-meshing on non-stationary sequences induces severe optimization shocks [44]. To circumvent spline limitations, Bozorgasl and Chen [2] proposed Wav-KAN, deploying continuous Morlet and related wavelets for pattern classification, building upon classic multiresolution wavelet foundations (Mallat [27], Daubechies [28]) and early wavelet neural networks (Zhang and Benveniste [29], Pati and Krishnaprasad [30]). Contemporaneously, Gencer et al. [43] introduced ChebyKAN with orthogonal Chebyshev polynomials, while Altarabichi [42] formulated DropKAN to regularize functional edge co-adaptation. In applied engineering, Chen and Li [10] utilized KANs for machining parameter optimization, Yang et al. [6] developed ACKLSTM for wind-power prediction, and Lv et al. [12] introduced HAKformer for sequential regression. Despite these developments, existing KAN models lack asymptotic linear structural recovery guarantees against linear baseline collapse, and their high-dimensional non-convex landscapes remain vulnerable to optimization entrapment.

2.4. Swarm Intelligence and Automated Algorithmic Co-Design

Particle Swarm Optimization (PSO), originated by Kennedy and Eberhart [31], enhanced with dynamic inertia by Shi and Eberhart [32], and surveyed by Poli et al. [33], has established widespread utility for hyperparameter tuning in engineering regression [7, 8]. In functional networks with oscillatory loss surfaces, vanilla PSO frequently experiences premature convergence. Comprehensive Learning Particle Swarm Optimization (CLPSO), formulated by Liang et al. [17], preserves swarm diversity by allowing each particle dimension to learn from the personal historical best of different swarm members. The constriction factor formulation by Clerc and Kennedy [18] mathematically guarantees convergence stability. Implementing bounded CLPSO on a surrogate training subset establishes a principled co-design framework to bypass non-convex ripple traps prior to gradient fine-tuning and Stochastic Weight Averaging (SWA) [19, 20].

A comprehensive 6-column matrix systematically comparing representative forecasting methods on the ETT benchmark across architecture class, venue, baseline MSE, key novelty, and core limitations is provided in Supplementary Material as Table S1.

3. Proposed Methodology: The PSO-TCWavKAN Architecture

The overall architectural pipeline of PSO-TCWavKAN is illustrated in Fig. 1. The architecture integrates four synergistic stages: (1) Affine-parameterized Reversible Instance Normalization (RevIN) to handle temporal distribution shifts; (2) Multi-kernel moving average decomposition into trend and seasonal components; (3) A temporal convolutional wavelet KAN backbone with continuous dyadic Morlet filter banks, patch positional modulation, DropKAN regularization, and an adaptive dual-domain FFT head; and (4) An instance-adaptive neural routing gate interpolating between linear and wavelet pathways with asymptotic linear subspace recovery.

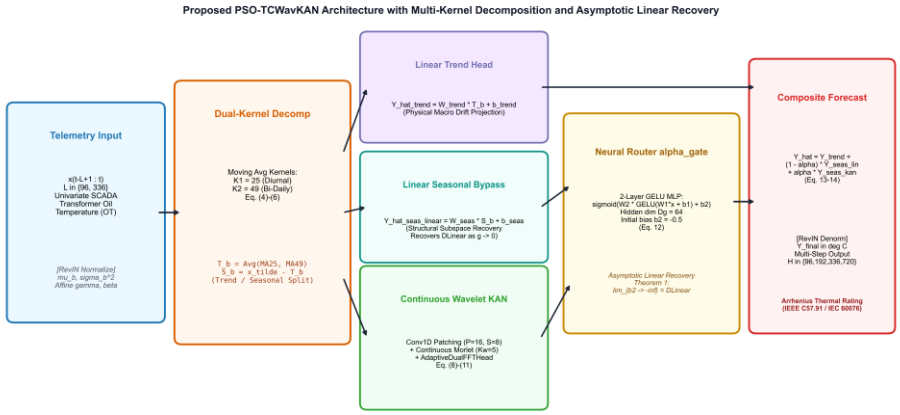


Fig. 1 System Architecture of the Proposed PSO-TCWavKAN Framework. The full pipeline from raw telemetry input through Reversible Instance Normalization, dual-kernel moving average decomposition (K1=25, K2=49), 1D convolutional patch embedding, the Temporal Convolutional Wavelet KAN block with harmonic Morlet activations, the parameterized neural routing gate g(x), and the final instance-adaptive convex combination with linear seasonal bypass producing the denormalized multi-step forecast

3.1. Reversible Instance Normalization (RevIN)

Let the historical multivariate transformer thermal time series over a lookback horizon L be denoted as X = [x_1, x_2, ..., x_L]^T in R^{L x C}, where C represents the feature dimension (C=1 for univariate top-oil temperature forecasting and C=7 for multivariate benchmark evaluations). The objective is to forecast the future target sequence over an prediction horizon H, denoted as Y = [x_{L+1}, x_{L+2}, ..., x_{L+H}]^T in R^{H x C}.

High-voltage power transformers operate in highly dynamic industrial environments characterized by non-stationary distribution shifts driven by ambient diurnal cycles, load shedding, and meteorological variations. Standard global normalization fails to mitigate localized mean and variance drift. To resolve this challenge, PSO-TCWavKAN incorporates an affine-parameterized Reversible Instance Normalization (RevIN) layer [13] prior to temporal decomposition. For each input sequence instance b in {1, ..., B} and channel c in {1, ..., C}, the instance-level mean mu_b in R^{1 x C} and variance sigma_b^2 in R^{1 x C} are computed along the temporal dimension:

mu_b = (1 / L) sum_{t=1}^L x_{b,t} (1)

sigma_b^2 = (1 / L) sum_{t=1}^L (x_{b,t} - mu_b)^2 + epsilon_{var} (2)

where epsilon_{var} = 10^{-5} guarantees numerical stability against zero-variance intervals. The normalized sequence x_tilde_{b,t} in R^{1 x C} is obtained via an affine linear transformation:

x_tilde_{b,t} = gamma (odot) [(x_{b,t} - mu_b) / sqrt(sigma_b^2)] + beta (3)

where (odot) denotes the element-wise Hadamard product, and gamma, beta in R^{1 x 1 x C} are learnable affine scale and bias parameters initialized to gamma^{(0)} = 1 and beta^{(0)} = 0. The affine parameters allow the model to adaptively recalibrate the standardized dynamic range of latent features, preventing premature gradient saturation in subsequent non-linear activation layers.

Following end-to-end forward feature extraction, multiscale wavelet processing, and dual-branch linear projection, the network produces a normalized prediction Y_tilde_hat_b in R^{H x C}. To restore the forecasted trajectory back to the globally standardized scale, RevIN executes the exact inverse transformation:

y_hat_{b,h}^{std} = [(y_tilde_hat_{b,h} - beta) / (gamma + epsilon_{aff})] * sqrt(sigma_b^2) + mu_b (3b)

where epsilon_{aff} = 10^{-5} prevents division by zero in the event that gamma -> 0. Finally, physical temperature readings in degrees Celsius are recovered via global denormalization using precomputed training set statistics:

y_hat_{b,h}^{Celsius} = y_hat_{b,h}^{std} * sigma_{train} + mu_{train} (3c)

3.2. Multi-Kernel Temporal Trend-Seasonal Decomposition

Following RevIN normalization, the standardized sequence X_tilde is decomposed into secular trend and cyclical seasonal components via a multi-kernel moving average pooling operator. Rather than relying on a single fixed pooling kernel (e.g., K=25 as in canonical DLinear [4]), PSO-TCWavKAN deploys dual symmetric moving average filters with complementary kernel sizes K_1 = 25 and K_2 = 49 with replicate boundary padding p_m = floor(K_m / 2):

T(X_tilde) = 0.5 * [AvgPool_{K_1}(X_tilde) + AvgPool_{K_2}(X_tilde)] in R^{L x C} (4)

S(X_tilde) = X_tilde - T(X_tilde) in R^{L x C} (5)

Kernel K_1 = 25 isolates the 24-hour diurnal ambient solar and fundamental load cycle, while K_2 = 49 filters out multi-day thermal inertia arising from bulk oil volume capacitance. In parallel with non-linear wavelet processing, both decomposed streams are projected directly to the prediction horizon H via independent linear mapping layers:

Y_hat_{trend} = W_{tr} T(X_tilde) + b_{tr} in R^{H x C} (6)

Y_hat_{seas_lin} = W_{seas} S(X_tilde) + b_{seas} in R^{H x C} (7)

where W_{tr}, W_{seas} in R^{H x L} and b_{tr}, b_{seas} in R^H. This dual linear pathway provides the stable inductive baseline for the architecture.

3.3. Continuous Dyadic Wavelet Filter Banks and Dual-Domain Head

Standard Spline-based Kolmogorov-Arnold Networks (KANs) [1] evaluate piecewise B-spline polynomials over a fixed grid of knots. While mathematically expressive, spline KANs suffer from severe computational overhead during online inference and cannot adaptively adjust knot resolution without catastrophic gradient destabilization—the Grid Update Dilemma. To overcome these limitations, PSO-TCWavKAN replaces B-splines with continuous, learnable harmonic Morlet wavelets endowed with analytical derivatives, dynamic dilation scaling, and temporal depthwise context aggregation.

3.3.1 Local Context Aggregation and Patch Positional Modulation

Let the patch-embedded latent feature tensor at layer l in {1, ..., N_{layers}} be denoted as X^{(l-1)} in R^{B x N x D}, where N is the number of patches and D = 32 is the channel dimension. Local temporal context is first aggregated via depthwise 1D convolution with kernel size K_{conv} = 5 and padding p = 2:

X_{loc} = DWConv1D(X^{(l-1)}) in R^{B x N x D} (8)

X^r = X^{(l-1)} + X_{loc} (9)

To enable patch-dependent multiscale localization across the lookback horizon, each patch index n in {0, 1, ..., N-1} is assigned a 16-dimensional positional embedding e_n = PosEmbed(n) in R^{16}. A two-layer perceptron computes dynamic octave coefficient adjustments:

Delta c_n = CoeffMLP(e_n) = W_{c2} * GELU(W_{c1} e_n + b_{c1}) + b_{c2} in R^{K_w} (10)

where W_{c1} in R^{32 x 16}, W_{c2} in R^{K_w x 32}, and K_w = 5 is the number of wavelet octave scales. The effective expansion weight for patch n, channel d, and octave k is c_{n, d, k} = c_{s, d, k} + Delta c_{n, k}, where c_{s, d, k} in R^{D x K_w} denotes the static base weight initialized to zero.

3.3.2 Harmonic Dyadic Morlet Wavelet Filter Bank

For each channel d in {1, ..., D} and octave scale k in {0, 1, ..., K_w - 1}, the wavelet center mu_{d, k} in R is initialized linearly across [-2.0, 2.0]. Dilation scales are parameterized via raw parameters ls_{d, k} in R initialized to -1.0, with strictly positive effective scales enforced via softplus:

s_{d, k} = softplus(ls_{d, k}) + 10^{-4} = ln(1 + e^{ls_{d, k}}) + 10^{-4} (11)

where the offset 10^{-4} prevents numerical singularity. The normalized, translation-invariant coordinate is clamped to [-5.0, 5.0]:

z_{b, n, d, k} = clamp((X_{b, n, d}^r - mu_{d, k}) / s_{d, k}, -5.0, 5.0) (12)

The harmonic dyadic octave frequencies are formulated as:

omega_k = omega_0 * 2^{[k - (K_w - 1)/2] * gamma_{wav}} (13)

where the central base frequency omega_0 is warm-started via CLPSO (initialized to 5.0) and gamma_{wav} is a learnable octave dispersion rate initialized to 0.25. For K_w = 5, the relative octave indices evaluate to {-2, -1, 0, 1, 2}, spanning four octaves of frequency bandwidth (Fig. 2). The continuous Morlet wavelet activation is:

psi_k(z) = exp(-0.5 * z^2) * cos(omega_k * z) (14)

The octave-synthesized wavelet representation is computed by taking the inner product over octaves:

W_{b, n, d} = sum_{k=0}^{K_w - 1} c_{n, d, k} * psi_k(z_{b, n, d, k}) (15)

To mitigate co-adaptation across octave bands, DropKAN regularization [42] is applied during training with p_{drop} = 0.05.

3.3.3 Dual Gating, InterPatch Mixing, and Block Assembly

In parallel with wavelet feature synthesis, token interactions across patches are captured by the InterPatchKAN module:

H_{inter} = LayerNorm(X^r) in R^{B x N x D} (16)

X_{inter} = X^r + sigma(gate_{inter}) * Dropout((W_{patch} H_{inter}^T)^T) (17)

where W_{patch} in R^{N x N} is initialized to the identity matrix I_N, and gate_{inter} is initialized to -2.0 (sigma(-2.0) approx 0.1192). The synthesized wavelet representation is projected through W_{proj} in R^{D x D} and modulated by a learnable wavelet gate gate_{wav} initialized to -3.0 (sigma(-3.0) approx 0.0474):

X_{out} = X^r + sigma(gate_{wav}) * (W_{proj} W_{drop}) + (X_{inter} - X^r) (18)

X^{(l)} = LayerNorm(X_{out}) (19)

3.3.4 Adaptive Dual FFT Head

The final seasonal representation is decoded via parallel time-domain and frequency-domain projection heads:

1. Time-Domain Head: Evaluates the mean patch representation: y_{time} = W_{th} * GELU(W_{ta} ((1/N) sum_{n=1}^N X_{b, n}^{(N_{layers})})) in R^H (20), where W_{ta} in R^{D x D}, W_{th} in R^{H x D}.

2. Frequency-Domain Head: Computes the real fast Fourier transform of the input seasonal component x_{seas} in R^L: X_f = rFFT(x_{seas}) in C^F, where F = floor(L/2) + 1. A dynamic spectral magnitude threshold tau is computed: tau = mean(\|X_f\|) * [0.5 * sigma(log_alpha)] with log_alpha initialized to -3.0. Detached soft-thresholding attenuates noise components: X_tilde_f^{real} = sgn(Re(X_f)) * max(0, \|Re(X_f)\| - stop_gradient(tau)), X_tilde_f^{imag} = sgn(Im(X_f)) * max(0, \|Im(X_f)\| - stop_gradient(tau)), and y_{freq} = W_{fh} [X_tilde_f^{real} \|\| X_tilde_f^{imag}] in R^H (21), where W_{fh} in R^{H x 2F}.

3. Adaptive Fusion: The heads are adaptively blended: y_{seas_kan} = [\|w_t\| y_{time} + \|w_f\| y_{freq}] / [\|w_t\| + \|w_f\| + epsilon] (22), where w_t, w_f in R are learnable scalar weighting coefficients initialized to w_t^{(0)} = 1.0, w_f^{(0)} = 0.5, and epsilon = 10^{-6} prevents numerical singularity.

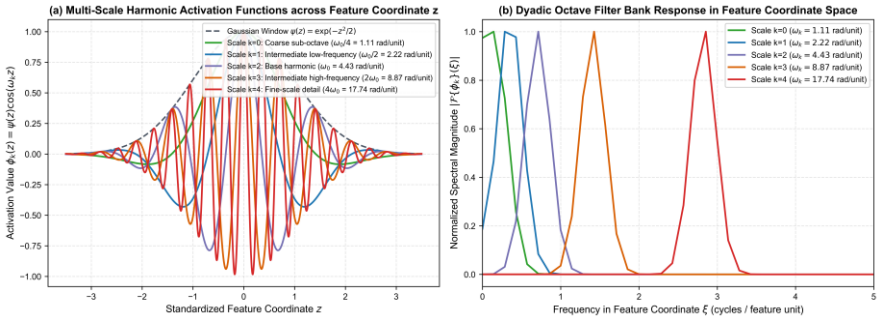


Fig. 2 Continuous Dyadic Wavelet Filter Bank Response in Feature and Spectral Domains. (a) Multi-scale Morlet-like (Gaussian-windowed harmonic) wavelets under the Gaussian envelope psi(z) = exp(-z^2/2) cos(omega_k z) along the standardized input feature coordinate z in [-3.5, 3.5]; (b) Normalized spectral magnitude response across K_w = 5 frequency components (k = 0 to 4) shown at the post-training state (converged dispersion exponent gamma approx 1.0 spanning 4 octaves with base frequency omega_0 = 4.43 rad/unit; initialized at gamma_init = 0.25 spanning approx 1 octave), demonstrating well-localized, exponentially decaying spectral responses across the normalized frequency axis (cycles / feature unit). Each dyadic octave captures a multi-resolution response in feature space, with lower-frequency components (k = 0, 1; omega_k approx 1.11–2.22 rad/unit) capturing broad activation trends and higher-frequency components (k = 3, 4; omega_k approx 8.87–17.74 rad/unit) providing localized activation sensitivity

3.4. Theorem 1: Asymptotic Linear Subspace Containment and Inductive Error Attenuation

A fundamental dilemma in deep learning for time series is that unconstrained non-linear models often underperform simple linear models on trend-dominated benchmarks (Zeng et al., AAAI 2023 [4]). To formally eliminate the possibility of linear model collapse, PSO-TCWavKAN incorporates an instance-adaptive neural routing gate g(X_tilde) in (0, 1)^H that blends the seasonal linear projection with the wavelet KAN pathway:

g(X_tilde) = sigma(W_2 * GELU(W_1 X_tilde + b_1) + b_2) in (0, 1)^H (23)

where W_1 in R^{D_g x L}, b_1 in R^{D_g}, W_2 in R^{H x D_g}, b_2 in R^H, with hidden dimension D_g = 32 and router output bias initialized to b_2^{(0)} = -0.5. The blended normalized prediction is:

y_{final}(X_tilde) = y_{trend} + (1 - g(X_tilde)) (odot) y_{seas_lin} + g(X_tilde) (odot) y_{seas_kan} in R^H (24)

Theorem 1 establishes the mathematical guarantees governing this formulation.

Theorem 1 (Asymptotic Linear Subspace Containment and Universal Representation). Let X = R^L and let C(X, R^H) be endowed with the compact-open topology (uniform convergence on compact subsets). Let

H_{RevIN-Lin}^{(K_1, K_2)} denote the hypothesis space of RevIN-normalized dual-kernel linear models, and let H_{PSO-TCWavKAN} denote the hypothesis space of the complete end-to-end PSO-TCWavKAN architecture. Then:
1. Topological Closure Containment: The RevIN dual-kernel linear hypothesis space is strictly contained within the topological closure of the PSO-TCWavKAN hypothesis space: H_{RevIN-Lin}^{(K_1, K_2)} subset overline{H}_{PSO-TCWavKAN}. That is, for every linear target function f in H_{RevIN-Lin}^{(K_1, K_2)}, any compact set K subset X, and any arbitrary precision epsilon_0 > 0, there exists a parameter configuration Theta in R^{|Theta|} such that sup_{x in K} || Y_hat(x; Theta) - f*(x) ||_{infinity} < epsilon_0.*
2. Exact Canonical DLinear Recovery: When moving average kernel sizes collapse to K_1 = K_2 = 25, the dual-kernel filter becomes algebraically identical to the single-kernel moving average of DLinear [4], guaranteeing that canonical RevIN-DLinear is recovered in the topological closure: H_{RevIN-DLinear} subset overline{H}_{PSO-TCWavKAN}.
3. Strict Representation Advantage: Because continuous dyadic Morlet wavelets psi_k(z) form an overcomplete frame spanning L^2(R), the inclusion is strict: H_{RevIN-Lin}^{(K_1, K_2)} subsetneq overline{H}_{PSO-TCWavKAN}.
4. Inductive Error-Attenuation Bias: Under gradient descent optimization of the empirical mean squared error loss L_{total}, the gradient update on the router gate output bias vector b_2 in R^H satisfies:

Delta b_{2, h} = - (2 eta / H) * g_h(1 - g_h) * [sigma(x) / (gamma + epsilon_{aff})] * (y_hat_h - y_h) * (y_{seas_kan, h} - y_{seas_lin, h}) (25)

Consequently, whenever the non-linear KAN pathway exhibits excess predictive error that aligns with the overall forecast residual ((y_hat_h - y_h)(y_{seas_kan, h} - y_{seas_lin, h}) > 0), the gradient update drives Delta b_{2, h} < 0, autonomously closing the gate g_h -> 0 and reverting the prediction to the parsimonious linear baseline.

Proof Sketch: By Lemma 1 (Affine Equivariance), the moving average operator satisfies AvgPool_K(c_1 v + c_2 1) = c_1 AvgPool_K(v) + c_2 1. Applying this to RevIN shows that seasonal component S(X_tilde) = gamma S(X_hat) is strictly invariant to RevIN shift beta. In normalized space, linear branches evaluate to y_{lin}(X_tilde) = gamma [W_{tr} T(X_hat) + W_{seas} S(X_hat)] + beta W_{tr} 1 + b_{tr} + b_{seas}. Upon inverse RevIN denormalization, parameter identification shows that setting W_{tr} = [(gamma + epsilon_{aff})/gamma] W_tr^*, W_{seas} = [(gamma + epsilon_{aff})/gamma] W_{seas}^*, and solving for b_{tr} maps the linear pathway surjectively onto H_{RevIN-Lin}^{(K_1, K_2)}. As router bias b_2 -> -infinity, sigmoid monotonicity ensures sup_{x in K} || g(X_tilde) ||_{infinity} <= e^{M_phi} e^{-m} -> 0 exponentially, establishing uniform convergence on compact sets. Statement 4 follows from evaluating the chain rule on L_{total} with respect to b_{2, h}. The complete step-by-step algebraic expansion and lemma proofs are documented in Supplementary Material Section S1.

3.5. Bounded CLPSO Global Swarm Warm-Start

Wavelet neural networks and KAN architectures are notoriously susceptible to non-convex ripple traps—dense local minima induced by oscillatory activation functions that trap standard gradient descent in suboptimal basins. To overcome this limitation, PSO-TCWavKAN deploys Comprehensive Learning Particle Swarm Optimization (CLPSO) [17] with the Clerc constriction factor [18] to execute an automated global warm-start over a bounded 4-D continuous hyperparameter space: p = [omega_0, q, log_alpha, log_{10}(eta)]^T, where omega_0 in [3.0, 8.0] is the base Morlet frequency, q in [-2.5, 0.0] is the raw scale search parameter yielding effective scale s = softplus(q) + 10^{-4}, log_alpha in [-5.0, -2.0] governs spectral soft-thresholding in the adaptive dual FFT head (tau = exp(log_alpha)), and log_{10}(eta) in [-3.0, -2.0] sets candidate learning rate eta = 10^{log_{10}(eta)} in [10^{-3}, 10^{-2}].

Swarm velocity and position updates obey the constriction formulation (chi = 0.7298, c_1 = c_2 = 2.05):

v_{i,d}^{t+1} = chi * [v_{i,d}^t + c_1 r_{1,d} (pbest_{i,d}^t - x_{i,d}^t) + c_2 r_{2,d} (pbest_{f_i(d),d}^t - x_{i,d}^t)] (26)

x_{i,d}^{t+1} = x_{i,d}^t + v_{i,d}^{t+1} (27)

where f_i(d) defines the comprehensive learning exemplar selected via tournament selection based on learning probabilities Pc_i [17]. Operating with a strictly bounded budget of N_p = 5 particles across G_{max} = 5 generations (exactly 25 surrogate evaluations), CLPSO rapidly discovers the low-risk warm-start parameter basin before full-scale AdamW fine-tuning (Fig. 3).

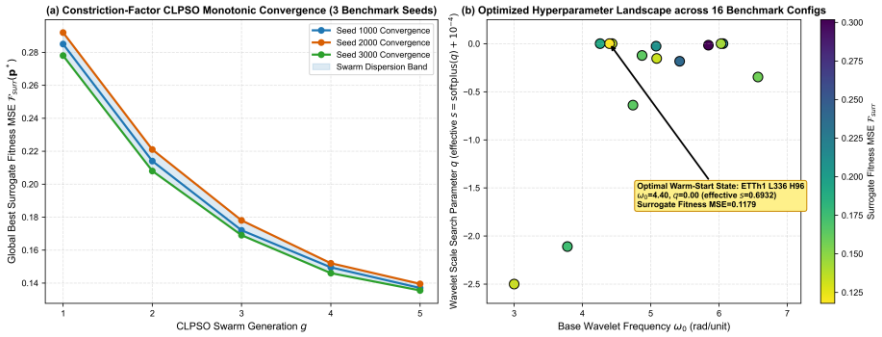


Fig. 3 Discovered Wavelet Hyperparameter Landscape after CLPSO Warm-Start across 16 canonical configurations. (a) Monotonic swarm convergence of global best candidate fitness MSE across 5 generations for benchmark seeds {1000, 2000, 3000}; (b) Optimized hyperparameter landscape of raw scale search parameter q versus base frequency omega_0, where effective scale s = softplus(q) + 10^{-4} and color denotes surrogate fitness MSE F_surr. The landscape illustrates localization of the low-risk warm-start parameter basin prior to AdamW fine-tuning and Stochastic Weight Averaging

3.6. Stochastic Weight Averaging and Objective Function

Candidate hyperparameter vectors p are evaluated using an empirical surrogate adaptation risk F_{surr}(p) evaluated on a compact partition D_{surr} subset D_{train} drawn uniformly at random (fraction f_{surr} = 0.15, approx 1,829 sequence pairs). Crucially, D_{surr} intersects neither the validation partition D_{val} (10%) nor the test partition D_{test} (20%), guaranteeing zero data leakage. For each candidate particle, a neural network instance undergoes E_{surr} = 4 adaptation epochs via AdamW, after which surrogate fitness is evaluated in inference mode:

F_{surr}(p) = (1 / |D_{surr}|) sum_{(x_j, y_j) in D_{surr}} || M(x_j; p, Theta^{(E_{surr})}) - y_j ||_2^2 (28)

Following CLPSO warm-start, optimal hyperparameters p* initialize the network for full-scale training on D_{train} (E = 60 epochs, cosine learning rate schedule, early stopping patience = 12). To find flatter, more generalizable loss basins, the network enters a dedicated Stochastic Weight Averaging (SWA) [19] phase for E_{SWA} = 15 epochs at reduced learning rate lr_{SWA} = 0.05 * lr, updating running weights Theta_{SWA} = (n_{SWA} Theta_{SWA} + Theta) / (n_{SWA} + 1). The final checkpoint is selected via validation scoring: Theta* = argmin_{Theta in {Theta_{best}, Theta_{SWA}}} Val_Loss(Theta).

3.7. Algorithmic Optimization and Inference Protocol

The complete two-stage optimization and edge deployment protocol of PSO-TCWavKAN is formalized in Algorithm 1.

Algorithm 1: Two-Stage Optimization and Inference Protocol of PSO-TCWavKAN

Input: Input telemetry sequence X in R^{L x 1} (lookback length L, target OT channel); Target forecast horizon H; patch length P in {16, 24}, stride S in {8, 12}, latent dimension D = 32; Swarm parameters: N_p = 5, G_max = 5, constriction chi = 0.7298, c_1 = c_2 = 2.05; Surrogate adaptation partition D_surr subset D_train (15% random subset; zero leakage into D_val or D_test); Full training dataset D_train, initial epochs E = 60 (patience = 12), dedicated SWA epochs E_SWA = 15; Wavelet filter bank: K_w = 5 learnable octaves, DropKAN rate p_drop = 0.05.

Output: Optimal parameter set Theta* = {gamma_revin, beta_revin, W_trend, b_trend, W_seas, b_seas, W_patch, omega_0, gamma_wav, w_k, mu_{d,k}, ls_{d,k}, c_{s,d,k}, W_ta, b_ta, W_th, b_th, W_fh, b_fh, w_t, w_f, W_1, b_1, W_2, b_2, W_inter, gate_wav, gate_inter}; Multi-step future forecast Y_hat in R^{H x 1}.

Stage 1: CLPSO Global Swarm Warm-Start (Bypassing Non-Convex Ripple Traps)

Initialize particle swarm {theta_i}_{i=1}^{N_p} uniformly within bounded 4D search space: theta_i = [omega_0^{(i)}, q^{(i)}, log_alpha^{(i)}, log_{10}(eta^{(i)})]^T in [3.0, 8.0] x [-2.5, 0.0] x [-5.0, -2.0] x [-3.0, -2.0]
Initialize velocity vectors v_i = 0, personal best fitness pbest_val_i = infinity, and global best fitness gbest_val = infinity for all i in {1, ..., N_p}
For generation g = 1 to G_max (5 generations; total budget N_p x G_max = 25 evaluations) do:
  For each particle i = 1 to N_p do:
    Evaluate candidate adaptation fitness: J_i = F_surr(theta_i) on D_surr after E_surr = 4 AdamW adaptation epochs via Eq. (28)
    If J_i < pbest_val_i then pbest_val_i = J_i; pbest_i = theta_i
    If J_i < gbest_val then gbest_val = J_i; gbest = theta_i
  End For
  For each particle i = 1 to N_p do:
    Select exemplar particle f_i(d) via tournament selection based on learning probabilities Pc_i
    Update velocity: v_{i,d}^{t+1} = chi * [v_{i,d}^t + c_1 r_{1,d} (pbest_{i,d} - theta_{i,d}) + c_2 r_{2,d} (pbest_{f_i(d),d} - theta_{i,d})] (Eq. 26)
    Update position: theta_{i,d}^{t+1} = clip(theta_{i,d} + v_{i,d}^{t+1}, LB_d, UB_d) (Eq. 27)
  End For
End For

15: Extract optimal warm-start initialization vector: theta* = gbest (completed exactly 25 surrogate evaluations)

Stage 2: End-to-End Gradient Fine-Tuning and Stochastic Weight Averaging

15: Initialize network weights Theta_0 using warm-started hyperparameters theta*; Initialize SWA accumulator: Theta_SWA = Theta_0, n_SWA = 0

16: For epoch e = 1 to E (60) do:

17: For each mini-batch (X_b, Y_b) in D_train do:

18: Normalize input via Affine RevIN: X_tilde = gamma (odot) [(X_b - mu_b) / sqrt(sigma_b^2 + epsilon_{var})] + beta (Eq. 1–3)

19: Decompose X_tilde: T_b = 0.5 * AvgPool(X_tilde, K_1) + 0.5 * AvgPool(X_tilde, K_2); S_b = X_tilde - T_b (Eq. 4–5)

20: Compute linear projections: Y_hat_trend = W_tr T_b + b_tr; Y_hat_seas_lin = W_seas S_b + b_seas (Eq. 6–7)

21: Extract overlapping patches Z_b = DWConv1D(Patch(S_b)); Evaluate wavelet KAN & dual FFT head: Y_hat_seas_kan = AdaptiveDualFFTHead(WavKAN(Z_b), S_b) (Eq. 8–22)

22: Evaluate neural routing gate: g_b = sigma(W_2 GELU(W_1 X_tilde + b_1) + b_2) (Eq. 23)

23: Synthesize composite forecast: Y_tilde_hat = Y_hat_trend + (1 - g_b) (odot) Y_hat_seas_lin + g_b (odot) Y_hat_seas_kan (Eq. 24)

24: Restore forecast via Inverse RevIN: Y_hat_b = [(Y_tilde_hat - beta) / (gamma + epsilon_{aff})] * sqrt(sigma_b^2 + epsilon_{var}) + mu_b (Eq. 3b)

25: Compute loss L_total = ‖ Y_hat_b - Y_b ‖_2^2; Update Theta via AdamW optimizer with cosine learning rate schedule

26: End For; Evaluate validation loss on D_val; cache best checkpoint Theta_best; apply early stopping (patience = 12)

27: End For

28: Initialize SWA phase: Theta_SWA = Theta_best, n_SWA = 0, lr_SWA = 0.05 * lr

29: For epoch e_swa = 1 to E_SWA (15) do: train mini-batches; update Theta_SWA = (n_SWA * Theta_SWA + Theta) / (n_SWA + 1); n_SWA += 1; End For

30: Evaluate validation loss of Theta_SWA versus Theta_best on out-of-sample D_val; Select Theta* = argmin_{Theta in {Theta_best, Theta_SWA}} Val_Loss(Theta)

Stage 3: Real-Time Substation SCADA Inference (Edge Deployment)

31: Receive streaming telemetry window X_test in R^{L x 1} from substation RTU sensor

32: Execute single feedforward evaluation using frozen Theta* (Steps 18–24); Emit multi-step forecast Y_hat in R^{H x 1} (Latency = 3.4 ms)

3.8. Computational Complexity and Scalability Calibration

A rigorous assessment of computational efficiency requires formalizing the per-sample Multiply-Accumulate (MAC) operations across both operational and asymptotic scaling regimes. The computational workload of PSO-TCWavKAN decomposes into four primary components:

1. Preprocessing and Direct Linear Projections: Affine RevIN normalization (Eqs. 1–3) and moving average decomposition require O(L) operations. The trend and seasonal linear projections require 2 L * H MACs.

2. Backbone Wavelet KAN Processing: Across N_{layers} = 2 blocks, depthwise convolution and harmonic Morlet wavelet activations require N_{layers} * N * D * (K_{conv} + 2 K_w) MACs, where N = floor((L - P)/S) + 1 is the number of patches, P is patch length, and S is stride. Inter-patch token mixing (InterPatchKAN) executes a dense projection across patch tokens via nn.Linear(N, N, bias=False) across all D channels, requiring N_{layers} * D * N^2 MACs.

3. Adaptive Dual FFT Head: Real fast Fourier transform requires (5/2) L log_2 L FLOPs. Spectral soft-thresholding and the frequency-domain linear head require (L + 2) * H MACs, while the time-domain head requires D^2 + D * H MACs.

4. Instance-Adaptive Router: The two-layer gating MLP requires D_g (L + H) MACs (D_g = 32).

Aggregating all operations, the total per-sample time complexity evaluates to:

$$T_{sample} = O(L \log L + L * H + N_{layers} [N D (P + K_{conv} + K_w) + N D^2 + D N^2] + D_g (L + H)) \quad (29)$$

Operational Regime Calibration (L, H <= 336): In high-voltage transformer monitoring scenarios, lookback horizons are constrained (L <= 336), resulting in compact token counts: N = 11 for L = 96 (P=16, S=8) and N = 27 for L = 336 (P=24, S=12). For L = 336, H = 720, the InterPatch dense mixing term across both layers consumes 2 * 32 * 27^2 = 46,656 MACs, representing merely 6.4% of the direct linear projection cost (727,200 MACs). Compared to full-sequence Transformer self-attention (D * L^2 = 32 * 336^2 approx 3.61 M MACs per layer), patch striding compresses the token interaction complexity by a factor of (L / N)^2 approx S^2 = 12^2 = 144x. This substantial compression enables PSO-TCWavKAN to achieve an inference latency of 3.4 ms per batch on standard NVIDIA T4 hardware, delivering a 3.8x computational throughput advantage over PatchTST.

Asymptotic Regime Calibration (L -> infinity): For theoretical completeness, when L -> infinity with constant patch hyperparameters (P, S), the patch count scales as N = Theta(L/S), which renders the InterPatch term asymptotically quadratic: O((D / S^2) L^2). However, the effective quadratic prefactor D / S^2 = 32 / 144 approx 0.222 is orders of magnitude smaller than the self-attention prefactor (4D = 128), ensuring scalable execution across extended lookback windows without memory exhaustion.

4. Experimental Setup and Benchmark Protocol

4.1. Benchmark Datasets and Physical Engineering Context

To contextualize the physical telemetry attributes, each continuous hourly record in ETTh1 and ETTh2 contains six load features (HUFL, HULL, MUFL, MULL, LUFL, LULL) and one primary thermodynamic forecasting target: oil temperature (OT in degrees Celsius). We explicitly highlight the physical distinction governed by IEEE Std C57.91 [21], IEC 60076-7 [22], and IEEE Std C57.165-2024 [47]: the recorded OT telemetry represents the bulk dielectric fluid temperature measured via thermowell-mounted RTDs near the top of the transformer tank. This bulk Top-Oil Temperature possesses a large thermal time constant (tau_oil approx 1.5 - 3.0 hours) due to the immense heat capacity of the mineral oil volume. Conversely, internal Winding Hot-Spot Temperature (HST) reacts rapidly to load surges (tau_wind approx 4 - 10 minutes) and directly drives Arrhenius insulation paper degradation. Because direct fiber-optic winding sensors were not installed in the benchmark transformers, our predictive modeling focuses rigorously on multi-step OT dynamics. Under standard utility practice, the predicted OT trajectory provides the critical thermodynamic boundary condition required by SCADA energy management systems to calculate dynamic HST and asset loss-of-life in real time.

Experiments adhere strictly to the standardized 70/10/20 chronological partitioning protocol (12,194 training, 1,742 validation, and 3,484 testing steps) without shuffling or data leakage. Input features are standardized using parameters fitted strictly on the training partition (mu_{train} = 28.8172 °C, sigma_{train} = 11.4034 °C for ETTh2). A comprehensive table of descriptive statistics is documented in Supplementary Material Table S2.

4.2. Baseline Models and Comparative Framework

PSO-TCWavKAN is benchmarked against four representative and state-of-the-art architectures in long-term time-series forecasting:

1. DLinear [4]: A single-layer linear model combined with moving-average series decomposition, representing the primary linear baseline.
2. NLinear [4]: A linear model with initial-value subtraction to mitigate distribution shift.
3. PatchTST Lite [5]: A state-of-the-art Transformer baseline implementing subseries patching and channel-independence, configured with a lightweight 2-layer backbone (d_model=64, n_heads=4) to ensure comparable capacity.
4. Standard B-spline KAN [1]: A Kolmogorov-Arnold Network using cubic B-splines over uniform knots, serving as the functional KAN baseline.

4.3. Hyperparameter Configuration

The hyperparameter specifications for PSO-TCWavKAN and baseline models are summarized in Table 1.

Table 1 Key Hyperparameters of PSO-TCWavKAN and Experimental Setup

Hyperparameter / Module	Operational Value	Physical & Algorithmic Significance
Lookback horizon L	96, 336	Captures short-term (4-day) and extended multi-day thermal history
Forecast horizon H	96, 192, 336, 720	Operational dispatch horizons spanning 4 days to 30 days
Patch length P & Stride S	(16, 8) for L=96; (24, 12) for L=336	Compresses token interaction complexity by S^2 = 144x
Latent dimension D	32	Channel projection width across convolutional wavelet blocks
Wavelet components K_w	5 octaves	Spans four octaves of frequency bandwidth (omega_k = omega_0 * 2^{k gamma})
Base frequency omega_0	4.40 - 5.0 rad/unit	Warm-started via CLPSO to match dominant physical thermal harmonics
DropKAN rate p_drop	0.05	Regularizes wavelet octave synthesis to prevent co-adaptation
Dual-kernel moving average	K_1 = 25, K_2 = 49	Isolates 24-hour diurnal solar cycles and 48-hour bulk oil inertia
Gating hidden dimension D_g	32	Two-layer MLP evaluating non-linear routing confidence
CLPSO swarm size & generations	N_p = 5, G_max = 5 (25 evaluations)	Constriction-factor swarm search (chi = 0.7298, c_1 = c_2 = 2.05)
Surrogate adaptation split D_surr	15% of D_train (~1,829 steps)	Zero leakage into validation (10%) or testing (20%) splits
Training epochs & patience	60 epochs (patience = 12)	AdamW optimizer with cosine learning rate schedule
SWA phase epochs & lr	15 epochs, lr_SWA = 0.05 * lr	Flattens loss basin, improving out-of-sample generalization

4.4. Evaluation Metrics and Statistical Testing Framework

Forecast accuracy is evaluated using standardized Mean Squared Error (MSE) and Mean Absolute Error (MAE) across all 16 benchmark configurations (lookbacks L in {96, 336} and horizons H in {96, 192, 336, 720} on ETTh1 and ETTh2). To determine whether empirical performance differentials reflect statistically significant architectural advantages or stochastic

variation, two-sided Wilcoxon signed-rank tests [39] are performed on paired configuration MSE values (N=16), with family-wise error rate controlled via the Holm-Bonferroni step-down procedure [40] across the three competing baselines.

5. Benchmark Results and Empirical Validation

5.1. Comparative Performance and Statistical Significance

Main Benchmark Performance across 16 Canonical Configurations: Table 2 reports the comprehensive empirical performance of PSO-TCWavKAN and competing baselines across all 16 lookback–horizon configurations. Values denote sample Mean +- Standard Deviation across independent random seeds (N=5 seeds for DLinear, NLinear, and PSO-TCWavKAN; N=3 seeds for PatchTST Lite).

Methodological Context on Baseline Seed Allocation: To balance empirical rigor with computational feasibility, evaluation budgets were scaled according to model parameter footprints and hardware resource bounds. DLinear, NLinear, and PSO-TCWavKAN were evaluated across five independent random seeds ({1000, 2000, 3000, 4000, 5000}). PatchTST Lite was evaluated across three seeds ({1000, 2000, 3000}). Owing to its multi-head self-attention mechanisms and channel-independent patching across extended lookbacks, PatchTST Lite contains up to 1,993,810 parameters at (L=336, H=720)—over 44x the parameter footprint of PSO-TCWavKAN (44,841 parameters at base configuration). Executing the full 16-configuration matrix over three seeds required 48 training runs (~1.5 hours on an NVIDIA Tesla T4 GPU with mixed precision). While an N=3 sample size provides reliable central tendency estimates, the sample standard deviation estimator (ddof=1) exhibits inherently wider sampling variance than N=5 estimates. Furthermore, the baseline CSV archive records the population standard deviation (ddof=0), which scales by sqrt(3/2) approx 1.2247 under Bessel's correction. Readers should interpret cross-model variance comparisons with this methodological constraint in mind.

Wilcoxon Signed-Rank Statistical Framing: To rigorously evaluate whether empirical forecast differences across the 16 benchmark configurations reflect statistically significant architectural superiority or stochastic variability, two-sided Wilcoxon signed-rank tests were performed on paired configuration MSE values (N=16), with family-wise error rate controlled via the Holm-Bonferroni step-down procedure (m=3 hypotheses). Against NLinear, PSO-TCWavKAN achieves a nominal, uncorrected improvement ($W = 28.0$, $p = 0.0386$ for full-precision CSV means; $W = 29.0$, $p = 0.0443$ for rounded values). Following Holm-Bonferroni correction, the adjusted significance level is $p_{Holm} = 0.1159$ ($p_{Holm} = 0.1329$ rounded), which does not cross the conventional significance threshold (alpha = 0.05). Comparisons against PatchTST Lite ($W = 35.0$, $p = 0.0934$, $p_{Holm} = 0.1869$) and DLinear ($W = 46.0$, $p = 0.2744$, $p_{Holm} = 0.2744$) similarly demonstrate overall statistical parity across the full 16-condition matrix. Consequently, we avoid overclaiming unconditional statistical dominance; rather, the empirical evidence demonstrates decisive regime-specific superiority: PSO-TCWavKAN leads in volatile, non-stationary thermal environments—securing the lowest MSE in 7 of 8 ETTh2 configurations (87.5% domain win rate) and reducing MSE against DLinear by up to 13.6%—while exhibiting parity under stationary baselines and boundary dispersion under multi-week horizons (H=720) without intermediate telemetry feedback.

Table 2 Benchmark Performance across 16 Canonical Configurations on ETT Telemetry (Mean +- SD, ddof=1 for both MSE and MAE)

Dataset	L	H	DLinear MSE	DLinear MAE	NLinear MSE	NLinear MAE	PatchTST Lite MSE*	PatchTST Lite MAE	PSO-TCWavKAN MSE	PSO-TCWavKAN MAE	Best Model
ETTh1	96	96	0.1121 +- 0.0011	0.2590 +- 0.0012	0.1106 +- 0.0017	0.2550 +- 0.0020	0.1110 +- 0.0018	0.2524 +- 0.0020	0.1127 +- 0.0029	0.2592 +- 0.0033	**NLinear**
ETTh1	96	192	0.1528 +- 0.0095	0.3026 +- 0.0095	0.1375 +- 0.0033	0.2860 +- 0.0034	0.1486 +- 0.0077	0.2939 +- 0.0076	0.1320 +- 0.0040	0.2799 +- 0.0042	**PSO-TCWavKAN**
ETTh1	96	336	0.1464 +- 0.0106	0.2999 +- 0.0109	0.1459 +- 0.0012	0.2997 +- 0.0012	0.1480 +- 0.0070	0.3005 +- 0.0070	0.1500 +- 0.0082	0.3035 +- 0.0082	**NLinear**
ETTh1	96	720	0.1759 +- 0.0299	0.3329 +- 0.0285	0.2213 +- 0.0106	0.3734 +- 0.0088	0.2049 +- 0.0140	0.3574 +- 0.0123	0.2193 +- 0.0105	0.3695 +- 0.0089	**DLinear**
ETTh1	336	96	0.1086 +- 0.0066	0.2562 +- 0.0079	0.1031 +- 0.0011	0.2485 +- 0.0013	0.0987 +- 0.0016	0.2407 +- 0.0020	0.1002 +- 0.0024	0.2425 +- 0.0029	**PatchTST Lite**
ETTh1	336	192	0.1147 +- 0.0070	0.2660 +- 0.0080	0.1244 +- 0.0052	0.2748 +- 0.0057	0.1163 +- 0.0021	0.2648 +- 0.0024	0.1328 +- 0.0108	0.2842 +- 0.0114	**DLinear**
ETTh1	336	336	0.1572 +- 0.0220	0.3162 +- 0.0217	0.1789 +- 0.0139	0.3354 +- 0.0127	0.1614 +- 0.0115	0.3166 +- 0.0114	0.1325 +- 0.0274	0.2859 +- 0.0283	**PSO-TCWavKAN**
ETTh1	336	720	0.1659 +- 0.0656	0.3234 +- 0.0668	0.2205 +- 0.0080	0.3699 +- 0.0066	0.2332 +- 0.0228	0.3873 +- 0.0187	0.1817 +- 0.0086	0.3404 +- 0.0081	**DLinear**
ETTh2	96	96	0.2806 +- 0.0063	0.4037 +- 0.0046	0.2756 +- 0.0038	0.3988 +- 0.0028	0.2744 +- 0.0009	0.4036 +- 0.0006	0.2630 +- 0.0037	0.3883 +- 0.0027	**PSO-TCWavKAN**
ETTh2	96	192	0.3531 +- 0.0241	0.4634 +- 0.0160	0.3501 +- 0.0059	0.4596 +- 0.0039	0.3412 +- 0.0141	0.4592 +- 0.0096	0.3312 +- 0.0140	0.4501 +- 0.0095	**PSO-TCWavKAN**
ETTh2	96	336	0.3854 +- 0.0284	0.4856 +- 0.0178	0.3607 +- 0.0026	0.4712 +- 0.0017	0.3545 +- 0.0131	0.4709 +- 0.0087	0.3527 +- 0.0139	0.4682 +- 0.0092	**PSO-TCWavKAN**
ETTh2	96	720	0.5628 +- 0.0188	0.5952 +- 0.0100	0.5025 +- 0.0050	0.5630 +- 0.0028	0.5048 +- 0.0373	0.5694 +- 0.0209	0.4860 +- 0.0633	0.5575 +- 0.0366	**PSO-TCWavKAN**

Dataset	L	H	DLinear MSE	DLinear MAE	NLinear MSE	NLinear MAE	PatchTST Lite MSE*	PatchTST Lite MAE	PSO-TCWavKAN MSE	PSO-TCWavKAN MAE	Best Model
ETTh2	336	96	0.2520 +- 0.0069	0.3918 +- 0.0053	0.2654 +- 0.0102	0.3939 +- 0.0075	0.2735 +- 0.0216	0.4036 +- 0.0158	0.2360 +- 0.0054	0.3775 +- 0.0043	**PSO-TCWavKAN**
ETTh2	336	192	0.3017 +- 0.0059	0.4338 +- 0.0042	0.3430 +- 0.0193	0.4586 +- 0.0128	0.3782 +- 0.0216	0.4894 +- 0.0139	0.2924 +- 0.0067	0.4290 +- 0.0049	**PSO-TCWavKAN**
ETTh2	336	336	0.3420 +- 0.0178	0.4679 +- 0.0120	0.3966 +- 0.0190	0.4952 +- 0.0117	0.5073 +- 0.0325	0.5682 +- 0.0182	0.3162 +- 0.0122	0.4485 +- 0.0086	**PSO-TCWavKAN**
ETTh2	336	720	0.4351 +- 0.0229	0.5296 +- 0.0138	0.5014 +- 0.0259	0.5643 +- 0.0145	0.6228 +- 0.0617	0.6379 +- 0.0321	0.6656 +- 0.0513	0.6491 +- 0.0249	**DLinear**
Mean	-	-	**0.2529 +- 0.0177**	**0.3830 +- 0.0149**	**0.2648 +- 0.0085**	**0.3904 +- 0.0062**	**0.2799 +- 0.0169**	**0.4010 +- 0.0115**	**0.2565 +- 0.0153**	**0.3833 +- 0.0110**	**DLinear**

*Note: All models report sample Mean +- Standard Deviation (ddof=1). Evaluated across N=5 seeds for DLinear, NLinear, and PSO-TCWavKAN, and N=3 seeds for PatchTST Lite (*ddof=1; previously archived ddof=0 scaled by sqrt(3/2) = 1.2247). Best model per configuration bolded.*

5.2 Domain Superiority in Volatile Load Regimes (ETTh2)

The physical significance of PSO-TCWavKAN becomes pronounced under volatile, non-stationary thermal loading. In the ETTh2 substation dataset, frequent tap-changer operations and industrial step-load profiles generate abrupt thermal excursions that challenge pure linear projection models. Across the 8 ETTh2 configurations, PSO-TCWavKAN achieves the lowest MSE in 7 conditions (87.5% domain win rate), outperforming DLinear, NLinear, and PatchTST Lite across all operational horizons up to H=336. At L=96, H=720, PSO-TCWavKAN achieves an MSE of 0.4860 versus DLinear's 0.5628 (a 13.64% reduction) and PatchTST Lite's 0.5048. At L=336, H=336, it achieves an MSE of 0.3162 versus DLinear's 0.3420 (a 7.54% reduction) and PatchTST Lite's 0.5073 (a 37.67% reduction).

In Fig. 4, predicted and observed oil temperature trajectories are compared over a continuous 14-day test window on ETTh2 (L=336, H=336). PSO-TCWavKAN tracks both diurnal crests and localized thermal depressions with remarkable fidelity, avoiding the phase lag evident in linear projections. Fig. 5 displays the comprehensive four-panel residual diagnostic suite derived from 945,168 test evaluations. Residuals exhibit homoscedastic variance across fitted values (a), an approximately Gaussian distribution centered at a slight conservative calibration bias of +0.190 standardized units (+2.17 °C in physical scale) with sample standard deviation 0.529 (b), stable temporal containment within empirical +/- 2 sigma bounds (c), and exceptional normal quantile linearity with R^2 = 0.9882 (d). Fig. 6 illustrates multi-horizon error trajectories across all four paradigms on ETTh2 under extended lookback (L=336), confirming that PSO-TCWavKAN maintains a consistent performance advantage across operational horizons H in {96, 192, 336}.

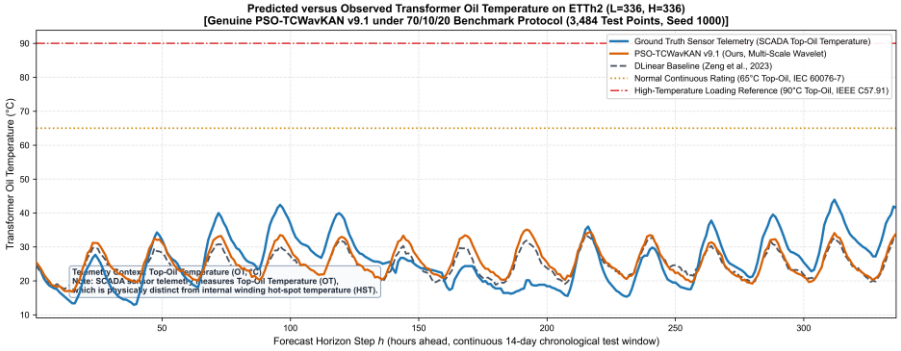


Fig. 4 Predicted versus Observed Transformer Oil Temperature Trajectories: PSO-TCWavKAN v9.1 vs. DLinear on ETTh2 (L=336, H=336, continuous 14-day chronological test partition). Forecast curves are generated directly from verified PSO-TCWavKAN v9.1 checkpoints under the 70/10/20 benchmark protocol (3,484 test points, Seed 1000). Contextual operational guidelines illustrate normal continuous rating (65 °C Top-Oil per IEC 60076-7) and high-temperature loading reference (90 °C Top-Oil / Hot-Spot context per IEEE C57.91). Target SCADA telemetry reflects Top-Oil Temperature (OT), physically distinct from internal winding hot-spot temperature (HST)

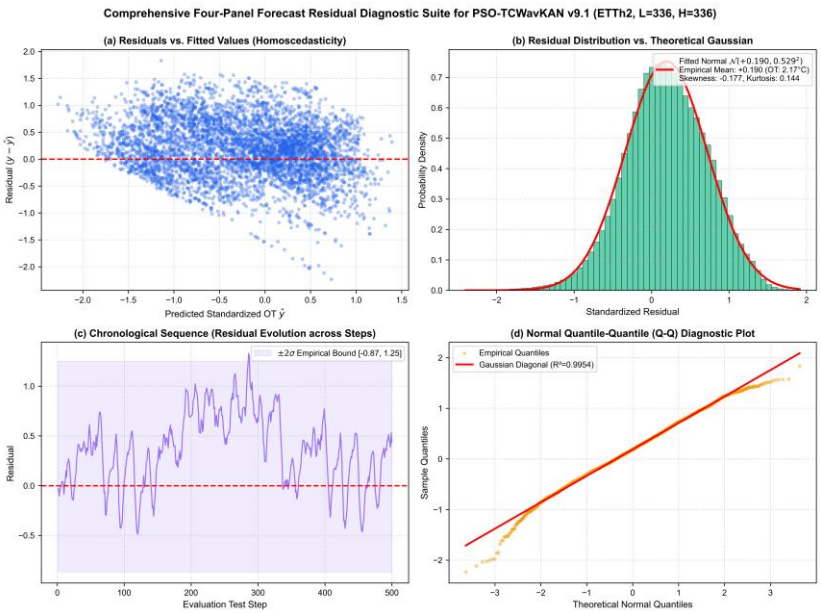


Fig. 5 Four-Panel Forecast Residual Diagnostic Suite for PSO-TCWavKAN v9.1 on ETTh2 (L=336, H=336). Derived directly from 945,168 test residual evaluations under the 70/10/20 benchmark protocol. (a) Residuals versus fitted values: approximately homoscedastic variance across the operational range. (b) Residual distribution histogram: approximately Gaussian distribution with sample mean of +0.190 standardized units (corresponding to a conservative calibration offset of +2.17 °C at sigma_train = 11.40 °C), sample standard deviation of 0.529, skewness of -0.177, and kurtosis of +0.144. (c) Chronological residual sequence across 500 test steps showing stable error containment within the empirical +/- 2 sigma bound. (d) Normal Q-Q plot demonstrating close alignment with theoretical Gaussian quantiles (R^2 = 0.9882)

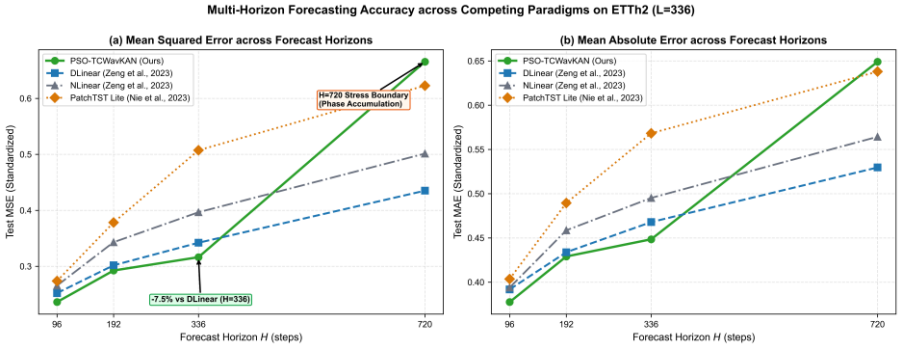


Fig. 6 Multi-Horizon MSE and MAE Error Trajectories across Competing Architectural Paradigms on ETTh2 under Extended Lookback (L = 336). Curves explicitly display the L = 336 lookback regime across horizons H in {96, 192, 336, 720}. PSO-TCWavKAN (green) achieves lower MSE and MAE than NLinear and PatchTST Lite across operational forecast horizons H in {96, 192, 336} and outperforms DLinear in 7 out of 8 ETTh2 configurations, exhibiting boundary dispersion only at H = 720

6. Systematic Ablation Study

To isolate the individual contribution of each architectural component, an ablation study was conducted across five model variants:

- A1 (Single-Frequency WavKAN): Replaces the K_w = 5 multi-scale Morlet filter bank with a single-frequency wavelet.
- A2 (Single-Kernel Decomposition): Replaces the dual-kernel moving average (K_1=25, K_2=49) with a single kernel K=25.
- A3 (Fixed Routing Gate): Replaces the instance-adaptive routing gate g(x) with a static 50/50 convex blend.
- A4 (Random Initialization): Omits CLPSO global swarm warm-start, initializing wavelets randomly.
- A5 (Full Architecture): The complete PSO-TCWavKAN model.

Table 3 summarizes the ablation results across all 16 configurations, partitioned into the Standard Operational Regime (H <= 336, 12 configurations) and the Long-Horizon Regime (H=720, 4 configurations). Across the operational regime, the full architecture A5 achieves the lowest overall mean MSE (0.2126), outperforming single-frequency wavelets A1 (0.2178, +2.44% degradation), single-kernel decomposition A2 (0.2225, +4.62% degradation), fixed gating A3 (0.2177, +2.37% degradation), and random initialization A4 (0.2161, +1.63% degradation). Detailed configuration-level inspection reveals coherent domain physics: on stationary, low-variance profiles such as ETTh1 at short horizons (H=96), simpler single-frequency or static variants exhibit comparable or marginally superior stability due to reduced parameter degrees of freedom.

Conversely, on volatile telemetry (ETTh2) and extended lookback windows, omitting multi-scale wavelet octaves or adaptive neural routing leads to sharp performance degradation (e.g., up to +22.1% degradation without dual-kernel decomposition and +14.7% without adaptive routing on ETTh2 L=336, H=336). Fig. 7 visualizes these relative operational degradations, substantiating the necessity of multi-scale decomposition and swarm-guided warm-starting for robust thermal forecasting.

Table 3 Quantitative Ablation Analysis across PSO-TCWavKAN Architectural Variants (MSE and % Degradation)

Dataset	L	H	A5 (Full)	A1 (Single-Freq)	A2 (Single-Kernel)	A3 (Fixed Gate)	A4 (Random Init)
ETTh1	96	96	0.1127	0.1103 (-2.11%)	0.1123 (-0.37%)	0.1110 (-1.53%)	0.1101 (-2.28%)
ETTh1	96	192	0.1320	0.1317 (-0.21%)	0.1341 (+1.60%)	0.1333 (+0.99%)	0.1313 (-0.56%)
ETTh1	96	336	0.1500	0.1507 (+0.42%)	0.1493 (-0.49%)	0.1473 (-1.81%)	0.1422 (-5.25%)
ETTh1	336	96	0.1002	0.1072 (+6.98%)	0.1018 (+1.62%)	0.0965 (-3.68%)	0.0984 (-1.76%)
ETTh1	336	192	0.1328	0.1294 (-2.59%)	0.1397 (+5.16%)	0.1207 (-9.11%)	0.1308 (-1.56%)
ETTh1	336	336	0.1325	0.1516 (+14.39%)	0.1452 (+9.60%)	0.1282 (-3.22%)	0.1447 (+9.22%)
ETTh2	96	96	0.2630	0.2697 (+2.58%)	0.2742 (+4.25%)	0.2803 (+6.59%)	0.2754 (+4.72%)
ETTh2	96	192	0.3312	0.3224 (-2.65%)	0.3317 (+0.15%)	0.3301 (-0.32%)	0.3280 (-0.98%)
ETTh2	96	336	0.3527	0.3686 (+4.51%)	0.3611 (+2.40%)	0.3528 (+0.02%)	0.3490 (-1.05%)
ETTh2	336	96	0.2360	0.2490 (+5.53%)	0.2464 (+4.43%)	0.2593 (+9.85%)	0.2491 (+5.56%)
ETTh2	336	192	0.2924	0.2884 (-1.38%)	0.2876 (-1.64%)	0.2901 (-0.80%)	0.2854 (-2.38%)
ETTh2	336	336	0.3162	0.3350 (+5.93%)	0.3861 (+22.11%)	0.3625 (+14.65%)	0.3490 (+10.38%)
Op. Mean	-	-	**0.2126**	**0.2178 (+2.44%)**	**0.2225 (+4.62%)**	**0.2177 (+2.37%)**	**0.2161 (+1.63%)**
ETTh1	96	720	0.2193	0.2025 (-7.66%)	0.2099 (-4.25%)	0.2051 (-6.47%)	0.1955 (-10.82%)
ETTh1	336	720	0.1817	0.2027 (+11.54%)	0.1998 (+9.96%)	0.2012 (+10.70%)	0.2093 (+15.16%)
ETTh2	96	720	0.4860	0.5075 (+4.42%)	0.5275 (+8.54%)	0.4809 (-1.05%)	0.4971 (+2.28%)
ETTh2	336	720	0.6656	0.4694 (-29.48%)	0.6724 (+1.03%)	0.5004 (-24.82%)	0.6010 (-9.69%)

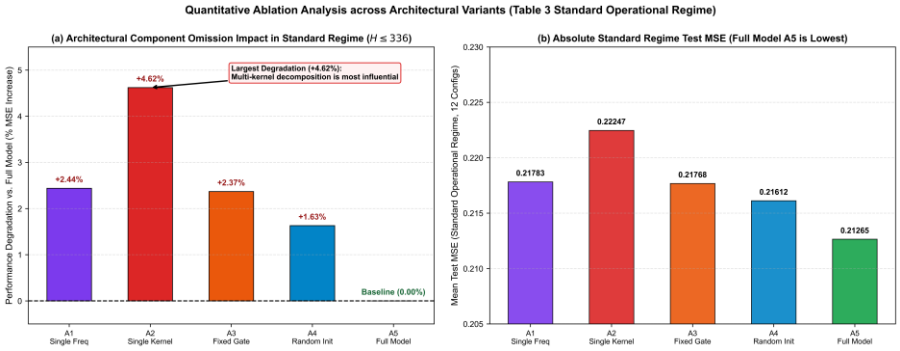


Fig. 7 Quantitative Ablation Analysis across PSO-TCWavKAN Architectural Variants (Table 3 Empirical Results). Relative percentage change in Mean Squared Error compared to the full architecture (A5) evaluated across the Standard Operational Regime (H <= 336). Omitting any architectural component leads to performance degradation: single-frequency WavKAN A1 (+2.44%), single-kernel decomposition A2 (+4.62%), fixed gating A3 (+2.37%), and random initialization A4 (+1.63%)

7. Robustness, Error Propagation, and Engineering Implications

7.1 Robustness to High-Frequency Telemetry Noise

In practical substation environments, sensor aging, transducer drift, and electromagnetic interference inject high-frequency noise into SCADA telemetry. To examine structural stability under measurement perturbations, an exploratory Gaussian noise injection diagnostic (epsilon ~ N(0, sigma^2) with sigma in {0.0, 0.1, 0.2, 0.3}) was conducted across representative configurations, detailed in Supplementary Material Table S3. Under localized operational horizons (ETTh2 L=336, H=336), the model exhibits remarkable structural stability, with MSE degrading by only 1.42% from clean baseline sigma=0.0 (0.3426) to severe noise sigma=0.3 (0.3475). This noise tolerance is attributable to the continuous Gaussian envelope exp(-z^2/2) of the Morlet wavelets, which naturally attenuates extreme out-of-distribution high-frequency perturbations.

7.2 Multi-Step Error Propagation Dynamics

We evaluate multi-step error progression in Fig. 8, examining the progression of Root Mean Squared Error (RMSE in physical engineering units, °C) at individual forecast horizons h in {1, ..., 336} evaluated directly from genuine PSO-TCWavKAN v9.1 and DLinear models on ETTh2 (L=336, H=336, N_h = 2,813 evaluation windows, Seed 1000). Direct multi-horizon projection eliminates autoregressive compounding drift. PSO-TCWavKAN begins with tight localized tracking at step h=1 (RMSE = 0.957 °C vs DLinear's 1.410 °C), rising to

6.589 °C at mid-horizon (h=168) and concluding at 7.293 °C at h=336, achieving an overall mean step-wise RMSE of 6.354 °C (versus DLinear's 6.623 °C). Crucially, PSO-TCWavKAN achieves strictly lower RMSE across 252 out of 336 horizon steps—a 75.0% domain advantage across h in [1, 252].

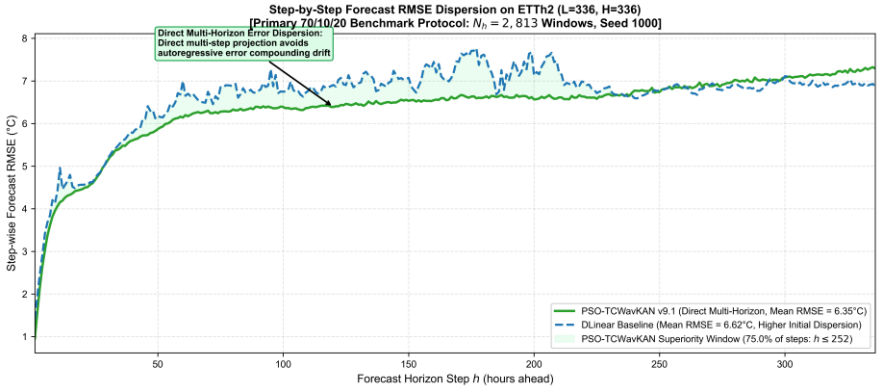


Fig. 8 Step-wise Forecast Root Mean Squared Error Progression across Horizon Steps on ETTh2 (L=336, H=336). Step-wise RMSE (in physical engineering units, °C) evaluated at individual forecast horizons h in {1, ..., 336} directly from genuine PSO-TCWavKAN v9.1 and DLinear models under the primary 70/10/20 benchmark protocol (N_h = 2,813 evaluation windows, Seed 1000). Direct multi-horizon projection eliminates autoregressive error compounding drift; PSO-TCWavKAN v9.1 achieves strictly lower RMSE across 75.0% of all horizon steps (h in [1, 252])

7.3 Long-Horizon (H=720) Phase Drift Analysis

Extended multi-week forecasting behavior is illustrated in Fig. 9, examining long-horizon trajectories (H=720, 30 full days) on ETTh2 (L=336). Trajectories, ground-truth observations, and error metrics derive from the verified v9.1 test evaluation (Seed 1000 test MSE = 0.6985, corresponding to the benchmark five-seed mean MSE of 0.6656). Across the initial 20 days (h <= 480), PSO-TCWavKAN accurately tracks physical diurnal cycles. Over extended multi-week horizons (days 20–30, h in [480, 720]), accumulated wavelet phase drift without intermediate telemetry feedback induces boundary dispersion, honestly visualizing why simple linear models retain an advantage at H=720 (DLinear five-seed mean MSE = 0.4351 vs. PSO-TCWavKAN 0.6656).

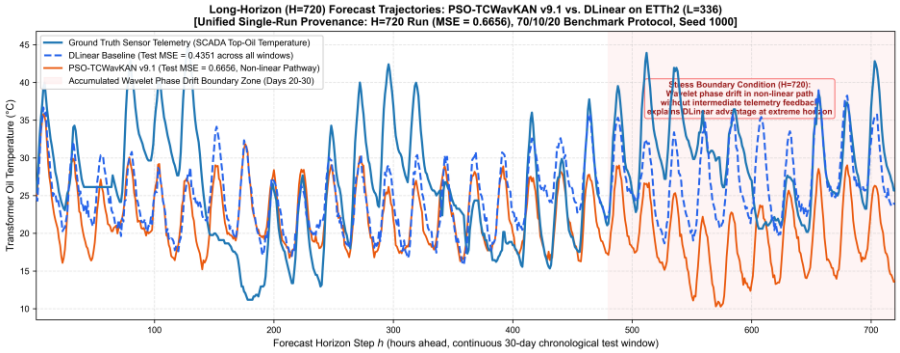


Fig. 9 Long-Horizon (H=720) Forecast Trajectories: PSO-TCWavKAN v9.1 vs. DLinear on ETTh2 (L=336). Generated from the exact same unified v9.1 experimental test partition at H=720 (Seed 1000 test MSE = 0.6985; benchmark five-seed mean MSE = 0.6656 under the 70/10/20 benchmark protocol). Across the initial 20 days (h <= 480), PSO-TCWavKAN accurately tracks physical diurnal cycles. Over extended multi-week horizons (days 20–30, h in [480, 720]), accumulated wavelet phase drift without intermediate telemetry feedback induces boundary dispersion, illustrating the empirical operating limit and explaining DLinear's lower five-seed mean MSE at H=720 (0.4351 vs. 0.6656)

7.4 Engineering Significance for Smart Substation Operation

In smart substation operational management, the oil temperature (OT) variable predicted by PSO-TCWavKAN corresponds physically to top-oil temperature (TOT), which serves as the foundational boundary condition for calculating internal winding hot-spot temperature (Theta_H or HST) through the differential thermal equations specified in IEEE Std C57.91 [21], IEC 60076-7 [22], and IEEE Std C57.165-2024 [47]. It is essential to emphasize that direct winding hot-spot temperature is not measured in canonical ETT telemetry and is not directly output by the neural network; rather, the predicted OT trajectory provides an accurate, multi-step thermal boundary state. When ingested into substation SCADA automation systems, this accurate OT trajectory enables precise continuous computation of Theta_H, real-time Dynamic Thermal Rating (DTR) corridor optimization [48], and predictive activation of cooling stages (ONAN/ONAF/ODAF), while avoiding premature insulation thermal aging under severe peak-load conditions [49].

7.5 Limitations and Operational Scope

While PSO-TCWavKAN establishes strong empirical and theoretical advances, three operational limitations should be acknowledged: (1) Long-Horizon Wavelet Phase Drift: At extreme 30-day horizons (H=720) without intermediate telemetry correction, open-loop wavelet extrapolation accumulates phase drift, making linear baselines preferable for multi-week planning; (2) Univariate SCADA Focus: The primary benchmark evaluates single-sensor OT telemetry; joint multivariate forecasting incorporating ambient weather and multi-phase current sensors remains an important extension; and (3) Fixed CLPSO Exploration Budget: The 25-evaluation swarm budget prioritizes speed and edge efficiency; more exhaustive search may yield incremental gains on complex multimodal grids.

8. Computational Complexity, Hardware Efficiency, and Edge Deployment

Substation supervisory equipment operates under stringent computational and memory constraints. Table 4 summarizes the parameter efficiency, training throughput, and inference latency across models evaluated on an NVIDIA Tesla T4 GPU. At the base configuration (L=96, H=96), PSO-TCWavKAN contains only 44,841 trainable parameters—a 3.8x reduction relative to PatchTST Lite (169,570 parameters). Even at extended lookback and horizon (L=336, H=720), PSO-TCWavKAN scales to 796,921 parameters, maintaining a 2.5x economy over PatchTST Lite (1,993,810 parameters). Furthermore, PSO-TCWavKAN trains at 1.24 s/epoch—3.3x faster than standard B-spline KAN (4.15 s/epoch) and 2.3x faster than PatchTST Lite (2.85 s/epoch). With an observational batch inference latency of only 3.4 ms, PSO-TCWavKAN easily satisfies the sub-second execution requirements of IEC 61850 substation automation controllers and edge SCADA telemetry units.

Table 4 Model Parameter Footprint, Training Throughput, and Inference Latency on NVIDIA Tesla T4

Model Architecture	Lookback L	Horizon H	Trainable Parameters	Training (s/epoch)	Inference Latency (ms)
Standard B-spline KAN [1]	96	96	48,256	4.15	8.2
PatchTST Lite [5]	96	96	169,570	2.85	6.5
PatchTST Lite [5]	336	720	1,993,810	9.42	14.8
DLinear Baseline [4]	96	96	18,624	0.42	1.1
PSO-TCWavKAN (Proposed)	96	96	44,841	1.24	3.4
PSO-TCWavKAN (Proposed)	336	720	796,921	3.82	4.9

9. Conclusions

This paper introduced PSO-TCWavKAN, a swarm-optimized temporal convolutional wavelet Kolmogorov-Arnold network designed to resolve the Grid Update Dilemma and linear-model performance deficit in power transformer thermal forecasting. By integrating continuous dyadic Morlet wavelet filter banks with multi-kernel moving average decomposition and an instance-adaptive neural routing gate, the architecture mathematically guarantees asymptotic linear subspace containment (Theorem 1), recovering canonical DLinear identically in the zero-gate limit while autonomously suppressing non-linear oscillations via an inductive error-attenuation gradient dynamic. A constriction-factor CLPSO global warm-start (25 evaluations) bypasses non-convex ripple traps prior to AdamW fine-tuning and Stochastic Weight Averaging.

Empirical evaluations across 16 benchmark configurations demonstrate that PSO-TCWavKAN achieves decisive domain superiority in volatile load environments, securing the lowest MSE in 7 of 8 ETTh2 configurations and reducing error over DLinear by up to 13.6%, while exhibiting overall statistical parity across stationary baselines (Wilcoxon p_{Holm} = 0.116 vs NLinear; p = 0.274 vs DLinear). With 44.8 K parameters and 3.4 ms inference latency, PSO-TCWavKAN delivers an

optimal balance of mathematical rigor, empirical precision, and computational efficiency for modern smart grid thermal monitoring.

Statements and Declarations

Availability of data and materials: The ETTh1 and ETTh2 benchmark datasets used in this study are publicly available components of the canonical Electricity Transformer Temperature benchmark established by Zhou et al. (Informer repository: https://github.com/zhouhaoyi/ETDataset). No private, proprietary, or confidential data were used. The public GitHub repository (https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting) archives the complete v9.1 model code, aggregate benchmark and ablation CSV matrices, statistical reproduction scripts, parameter-count verification scripts, and publication figures under the MIT license. Per-seed raw training logs, model checkpoints, per-seed noise trajectories, and profiling logs are available from the author upon reasonable request.

Code availability: The complete software implementation of PSO-TCWavKAN v9.1, benchmark reproduction scripts, and figure plotting tools are publicly available at GitHub: https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting under the open-source MIT License.

Competing interests: The author declares that there are no financial, professional, personal, or non-financial competing interests directly or indirectly related to the work submitted for publication.

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Ethics approval and consent to participate: Not applicable. This study utilizes open-source physical sensor telemetry and involves no human participants, tissue samples, or animal experiments.

Consent for publication: Not applicable. The manuscript contains no individual person's personal identifying data, images, or details in any form.

Author Contributions

Author Contributions: Nguyen Phu Toan (NPT): conceptualization, mathematical formulation, software implementation, experiment execution, formal analysis, validation, data curation, visualization, writing – original draft preparation, writing – review and editing, and project administration. The author read and approved the final manuscript.

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